THE IMPACT OF ONLINE FAN ENGAGEMENT ON MAJOR LEAGUE BASEBALL ATTENDANCE

by

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A Capstone Project
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Date: ____________________ Fall Semester 2025
George Mason University
Fairfax, VA

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Bachelor of Science
My Other Former School, Year of first degree

Director: Dr. Kent Miller, Professor
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Dedication

This article is dedicated to those who believe that data is more than just a technical tool or observation. Data is a lens through which we can better understand the patterns that drive human behavior—how communities form, evolve, and express their passion through sport.

This project is inspired by anyone who values studying not only what people do, but why they do it. Although online fan engagement might seem digital and, therefore irrelevant, behind every interaction lies a story of identity, belonging, and connection. This article is dedicated to those who recognize that these stories deserve to be explored with the same rigor as scientific inquiry. This work is dedicated to the scholars and mentors who taught me that this interest must be balanced with discipline, that statistical models can coexist with human meaning, and that evidence-based reasoning can help us understand even the most emotional aspects of life more deeply. Their guidance has shaped my belief that sports analytics is not just about numbers, but about people, culture, and the collective experience of fandom.

Above all, this capstone project is for all those who are inspired by the idea that the bonds formed between sports fans and spectators can be revealed through thoughtful analysis. I sincerely thank all those who constantly question, seek patterns, and believe in the promise of data-driven insights.

Acknowledgments

I would like to express my profound appreciation to all those who supported, guided, and inspired me throughout the development of this capstone project. This research is not only a milestone in academics, but also a personal journey shaped by the mentorship, encouragement, and intellectual support of many.

I would also like to express profound appreciation to Dr. Kent Miller, who led the project with rigor and curiosity. It was his willingness to question assumptions, explain ambiguities, and promote in-depth analysis that greatly helped to shape both the direction and quality of this research. I am grateful for his continued support and for creating an environment where questions are welcomed and exploration is encouraged. I am also very grateful to the faculty and staff in the Department of Computational and Data Science at George Mason University. Through teaching, feedback, and commitment to interdisciplinary thinking, they have strengthened the foundation on which this research was built. The lessons learned through classes, discussions, and practical analytical exercises have made a great impact on my growth as a researcher. To my colleagues and classmates, thank you for the conversations, challenges, and encouragements we shared in the process. The collaborative learning and mutual support have turned this journey into one that is not only productive but deeply meaningful. I would also like to thanks for the many people in the sports analytics community who inspired this project. Their work showed how data can shed light on the social and behavioral aspects of sports, and their contributions have encouraged me to explore further the area of online fan engagement.Last but not least, I would like to thank my family and friends, who believe in my capabilities and stand by me during those long nights of writing, modeling, and revising. Their support has given me the confidence to tackle this project with purpose and resilience.

I am grateful to everyone who has contributed directly or indirectly to this work, helping me reach this important milestone. Your influence will continue to be a strong force in my growth, not only as a researcher but also as a member of the analytics community.

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Abstract

THE IMPACT OF ONLINE FAN ENGAGEMENT ON MAJOR LEAGUE BASEBALL ATTENDANCE

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This project examines the relationship between online fan engagement and in-person attendance for Major League Baseball (MLB) teams. As digital platforms increasingly influence how fans interact with professional sports teams, understanding whether online activity leads to meaningful behavioral outcomes has become a crucial question for teams and analysts. This study examines how metrics such as social media interaction, team-produced online content, and digital fan engagement are associated with MLB regular season game-day attendance.

Combining publicly available MLB attendance records with search metrics collected from Google Trends, this analysis applies both descriptive and multiple regression modeling to assess the extent to which online engagement predicts attendance after accounting for traditional determinants such as team performance, market size, day/night game schedule, and opponent performance. This modeling approach verifies whether online fan behavior provides explanatory power beyond traditional predictors of attendance demand.

The results show that online fan engagement has a statistically significant and positive correlation with home game attendance, even after controlling for game performance and

contextual factors. Certain forms of engagement, such as highly interactive posts and real-time fan feedback, have stronger predictive power, suggesting that digital platforms can serve not only as marketing tools but also as behavioral signals that signal fan motivation and community activation.

These results highlight the growing importance of the digital ecosystem in shaping modern sports consumption and suggest that MLB teams could benefit from integrating online engagement analytics into attendance prediction and fan engagement strategies. This study contributes to a broader understanding of how virtual fan behavior complements traditional audience models and highlights the importance of digital interaction in sports analytics.

Chapter 1: Introduction

Fan attendance has long been a key performance indicator for Major League Baseball (MLB), influencing revenue, game atmosphere, and team strategy. Traditional determinants such as team performance, market size, star players, pricing, and game schedules have been widely studied and continue to play a significant role. However, as fan interaction increasingly takes place in the digital space, new forms of online behavior can provide additional insights into how interest develops and translates into in-person attendance. Among these digital signals, Google Trends provides a high-frequency measure of public interest that reflects shifts in fan curiosity and engagement in near real-time.

Despite the growing importance of digital engagement, empirical research examining whether online search interest serves as a short-term predictor of team-week MLB attendance is relatively scarce. If search behavior reliably precedes attendance changes, it could provide teams with valuable early indicators that complement existing predictive tools.

This project investigates whether Google Trends-based search intensity can explain weekly fluctuations in home attendance for MLB teams in the National League East. Using R, we constructed a team-week panel dataset based on MLB game logs and Google Trends data from the 2023-2025 regular season. After accounting for team performance and the number of home games per week, we applied two-way fixed-effects panel regression to estimate how lagged search interest correlates with subsequent attendance.

The purpose of this study was to integrate digital behavior into existing attendance models and evaluate the utility of search interest as a leading indicator of fan demand. As MLB teams continue to expand their digital engagement strategies, understanding how online interest correlates with actual attendance can provide practical value for forecasting, marketing, and fan experience planning.

Chapter 2: Prior Research

2.1 The Evolution of Traditional Attendance Determinants

Professional baseball attendance research has evolved over several generations, reflecting changes in the sport's structure, information environment, and consumer behavior. Early attendance models in sports economics overwhelmingly emphasized stadium performance as the primary determinant of attendance demand. In particular, winning percentage was closely correlated with team and season-to-season attendance changes, demonstrating the dominant performance-based explanation.

One of the most widely cited examples of this relationship is Glass's analysis of winning percentage and normalized attendance in Major League Baseball over several decades. [1]. Although the magnitude of the effect varied across franchises and markets, fan participation consistently increased when teams performed well and decreased when they lost. This pattern became a cornerstone of sports economics research and influenced modeling for decades to come.

However, over time, scholars began to recognize that team performance alone could not fully explain attendance fluctuations. The emergence of regional scale, modern stadiums, television broadcasts, and diverse entertainment options have created new factors that interact with traditional determinants. For example, regional scale has been observed to influence each team's baseline potential attendance, while stadium characteristics such as seating capacity, accessibility, and facility quality increasingly influence fan preferences.

Furthermore, the increasing commercialization of professional baseball throughout the twentieth century expanded the role of marketing, promotion, and strategic scheduling. Researchers began incorporating variables such as midweek versus weekend games, day versus night games, and special events. These additional factors demonstrated that attendance is not solely determined by win-loss records, but rather by a combination of structural and contextual factors.

With the advent of the digital age, new determinants have emerged. The proliferation of online content, social media, and real-time information flow has transformed how fans engage with baseball. Unlike in the past, today's fans watch baseball simultaneously across multiple channels, receive instant updates on team performance, and form emotional and narrative attachments through online discourse. These developments have made the relationship between fan interest and stadium attendance more dynamic.

Synthesizing the evolution of traditional audience determinants reveals that audience behavior is highly sensitive to changes in the broader media and technological environment. Although early research prioritized performance and market fundamentals, recent research demonstrates that audiences are shaped by complex economic, social, and behavioral factors. This expanded understanding provides a foundation for analyzing how digital information sources, such as online search interest, can influence today's audiences.

2.2 Economic Determinants of MLB Attendance

Another important strand of audience research focuses on how broader economic conditions affect fans' willingness and ability to attend games. Sports consumption, like many other entertainment activities, is discretionary in nature and therefore sensitive to changes in income, employment stability, and household finances.

Hong, Mondello, and Coates analyzed the impact of the 2008 financial crisis on MLB attendance, demonstrating the economic sensitivity of the league's demand structure.[2] Using weekly game data, they found that economic downturns were associated with a significant decline in fan participation. In particular, their prediction of a 6.5% decline in attendance closely matched the actual league-wide decline, strongly suggesting the explanatory power of macroeconomic variables.

These results support existing theories that economic downturns and financial uncertainty constrain household budgets, thus reducing discretionary spending, such as live sports. In addition, the impact of economic shocks varies between markets. Large metropolitan markets with large income bases tend to be relatively resilient, while smaller markets tend to be more exposed to economic fluctuations.

Previous studies highlight the influence of population size, metropolitan density, and the presence of competing franchises in the vicinity. While larger metropolitan areas tend to maintain stable attendance due to their broader fan bases, areas with two or more major league teams can experience cannibalization effects as entertainment spending is dispersed across multiple options.

The interplay of athletic performance, market size, economic conditions, and stadium characteristics demonstrates that attendance exists within a complex economic system that cannot be explained by a single factor. In other words, demand for baseball cannot be understood in isolation from the broader financial and demographic landscape.

This complexity further emphasizes the need to consider new sources of information, such as online search interest. Digital indicators react more quickly and reflect public sentiment more and provide additional value in explaining short-term attendance fluctuations even when traditional economic variables fluctuate relatively slowly.

2.3 Performance, Competitive Situation, and Stadium Effect

As mentioned earlier, team performance is one of the most studied factors that influence baseball attendance. The correlation between wins and attendance, documented by Glass [1], highlights the intuitive allure that fans feel for successful teams. While the performance effect persists in modern baseball, its magnitude varies across markets. Some teams, such as New York and Seattle, exhibit highly responsive attendance, while others exhibit more stable attendance regardless of win-loss records.

Beyond performance, the competitive context of each game significantly influences attendance. Rivalry matches, wild-card races, and series often generate attendance spikes, reflecting narrative tension that excites fans beyond mere wins and losses. Furthermore, the concept of competitive balance is a key factor in explaining fan demand, with research showing that fans tend to prefer matches with high uncertainty about the outcome.

Stadium characteristics also play a significant role in determining attendance. New stadiums tend to increase attendance in the years following their opening through the novelty effect. Qualitative elements of the stadium environment, such as seating comfort, visibility, amenities and quality of the concession stand, directly enhance the fan experience and significantly impact attendance. Recent research highlights the key role these stadium infrastructure and market factors play in shaping long-term MLB attendance. [3]

Stadium capacity also interacts with fluctuation in demand. In large stadiums, supply can exceed demand, diluting the impact of game performance. In contrast, smaller, modern stadiums, designed for intimacy and experience, maintain stable attendance even during fluctuating game performance.

These traditional factors are essential control variables in empirical models that assesses the impact of digital engagement. If online search interest simply reflects game performance or stadium quality, it is unlikely to have independent explanatory power. A key question this study addresses is whether digital metrics can capture short-term interest changes that are not fully explained by these traditional variables.

2.4 Digital Engagement and Online Attention in Sports

Digital behaviors are increasingly shaping how fans interact with professional sports. While previous generations obtained information through newspapers, radio, and television, modern fans follow their teams through digital platforms, search engines, and real-time updates. This shift has reduced the cost of accessing information, expanded the scope of what fans can monitor, and accelerated the pace of storytelling. Consequently, today's fans form impressions and make engagement decisions based on a continuous stream of digital signals, rather than intermittent or scheduled media updates. This shift has fundamentally altered the flow of information surrounding sports and created new experiential opportunities to measure and predict fan engagement, particularly through high-frequency data that captures short-term fluctuations in public attention.

Researchers studying digital engagement have examined various online behaviors across various sporting contexts. For example, Malagón-Selma and colleagues demonstrated that Google Trends effectively captures fluctuations in a football player's popularity, reflecting public awareness, media exposure, and narrative prominence in real time. [4]. Their findings suggest that search interest is not simply an indicator of player performance but reflects a variety of external factors, such as trade rumors, controversies, highlight reels, and increased broadcast exposure. This digital tracking complements traditional performance metrics by capturing broader social and cultural dimensions of fan interest that are difficult to observe through game statistics alone. The ability of search behavior to integrate emotional responses, media dynamics, and public discourse provides researchers with a richer understanding of how interest evolves throughout the season.

Mueller applied a random forest approach to baseball attendance prediction, demonstrating that nonlinear machine learning models can outperform traditional regression analysis by better capturing the interactions among determinants. [5]. His findings suggest that audience behavior is shaped by complex relationships among performance metrics, market characteristics, and scheduling features, patterns that can be difficult to detect with linear models. While his study did not include Google Trends, it highlights the growing importance of high-frequency, high-dimensional data in modern sports analytics. Moreover, Müller's approach demonstrates how flexible predictive models can uncover the hidden structure of fan behavior, suggesting that incorporating digital interest variables can further enhance model performance by capturing rapidly evolving aspects of consumer interest.

Digital engagement can mirror and influence offline fan behavior. In some cases, increased online engagement amplifies the effects of team performance, as increased search interest further reinforces fans' emotional investment and expectations. In other cases, digital engagement can partially replace offline attendance when fans feel sufficiently connected through online updates, highlight reels, or social media interactions. These two possibilities highlight that online and offline behaviors are not always complementary. The direction of the relationship can vary depending on team quality, market size, media coverage, or broader social factors. Therefore, understanding the nature of this online-offline relationship is essential, as digital engagement can increase or decrease stadium attendance depending on the situation. This ambiguity highlights the need for empirical analysis that considers the timing, context, and variation across fan segments.

Google Trends offers a particularly promising tool for studying these dynamics. Search data is collected daily or weekly, allowing us to capture short-term shifts in public interest triggered by trades, injuries, controversies, surprising performances, or new media stories. These spikes often occur before fan behaviors take effect, making Google Trends ideal for identifying early signals of audience fluctuations. Search data also reveals how quickly fans react to new information, providing insight into how quickly interest rises, wanes, or stabilizes after major events. By measuring real-time public curiosity and engagement, Google Trends enables researchers to analyze whether digital interest serves as a leading indicator of offline demand, offering a level of temporal precision unmatched by traditional performance or economic metrics.

2.5 Google Trends, Predictive Modeling, and Machine Learning

Google Trends has become a widely used instrument for measuring public interest, forecasting behavior, and analyzing information dynamics in near real time. Because it reflects the volume of queries submitted by millions of users, it offers a unique lens into how attention forms, rises, and dissipates across topics. Jun, Yoo, and Choi [6] trace the evolution of Google Trends from simple descriptive applications to more advanced predictive modeling frameworks, noting that search behavior increasingly serves as a proxy for collective attention across fields such as economics, marketing, and social science. Their work emphasizes that the value of Google Trends lies not only in the volume of searches but in the temporal patterns through which public curiosity and interest unfold.

The relevance of search data to sports analytics is multifaceted. First, it captures rapid fluctuations in fan awareness, allowing researchers to detect changes in interest that may not yet appear in game-level metrics or traditional surveys. Second, search data reflect narrative and emotional cycles such as excitement surrounding a trade, anxiety over an injury, or media-driven storylines that shape fan sentiment throughout a season. Third, because search behavior often precedes consumer decisions, it holds potential as a leading indicator of ticket purchases, merchandise spending, and ultimately game attendance. In this sense, search interest operates as an early signal of shifting motivation levels among fans.

Parallel evidence from other sports confirms the predictive utility of online search activity. Google Trends predicts soccer player transfer value with notable accuracy, suggesting that search interest captures dimensions of popularity, reputation, and narrative momentum not fully represented in performance statistics alone [4]. Their results imply that digital attention measures can reveal intangible factors that shape market outcomes in professional sports.

The broader machine learning literature also underscores the value of integrating digital variables into analytical frameworks. Mueller's work demonstrates that nonlinear models can successfully capture complex relationships among attendance determinants, especially when paired with high-frequency inputs [5]. His findings highlight that fan behavior is shaped by interactions between performance metrics, scheduling patterns, and contextual factors, patterns that may remain hidden in traditional models. These insights suggest that combining real-time search data with flexible machine learning techniques may substantially improve attendance forecasting by leveraging both structural and behavioral components of demand.

Google Trends, therefore represents a natural extension of prior attendance research. Its high temporal resolution aligns well with MLB's weekly scheduling structure, making it suitable for constructing panel datasets that capture short-term attention cycles and behavioral responses. Because search interest often reacts immediately to news events, while attendance decisions unfold over several days, it provides a unique opportunity to examine lead, relationships, and test whether digital engagement reliably precedes offline attendance decisions. This integration of digital attention metrics with traditional demand models offers a pathway to more precise, dynamic, and behaviorally informed analyses of fan engagement.

2.6 Online-to-Offline Dynamics and Theoretical Integration

Although MLB attendance research has traditionally focused on team performance, economic conditions, and stadium characteristics, studies from other disciplines provide valuable theoretical insights into how digital attention translates into offline behavior. Research in economics, marketing, and information systems consistently finds that search activity often correlates with demand for offline goods and services, although the direction and magnitude of this effect vary by context.

Examples include consumer markets where search interest predicts sales surges, financial markets where search intensity correlates with trading volume and volatility, and public health monitoring where search patterns provide early warnings of disease outbreaks. These findings demonstrate the broad applicability of search behavior as a behavioral signal.

Applied to sports, search behavior may function as an indicator of rising fan curiosity, information seeking, or emotional engagement. Fans often turn to search engines for news about roster moves, performance fluctuations, or major news events. As a result, spikes in search volume may capture attention cycles that precede attendance decisions.

At the same time, substitution effects are possible. In some settings, online engagement can satisfy fan interest without requiring in-person participation. This possibility underscores the need for empirical evaluation, rather than assuming a unidirectional or uniformly positive relationship between digital engagement and attendance.

2.7 Synthesis and Research Gap

Across the literature on traditional attendance determinants, macroeconomic influences, performance effects, digital engagement, and predictive modeling, several themes emerge. First, attendance is shaped by a complex interplay of structural, informational, and behavioral factors. Second, modern fans engage with baseball through digital environments that generate rich, high-frequency data. Third, Google Trends provides a real-time measure of public attention that may reveal short-term dynamics not captured in traditional datasets.

Despite these developments, research has not yet systematically evaluated whether Google Trends predicts MLB attendance at the weekly level. Existing studies typically rely on seasonal, yearly, or game-level data, with limited integration of digital trace variables such as search behavior. Moreover, few studies combine high-frequency attention measures with econometric models capable of isolating within-team and within-week variation. Given the panel structure of MLB scheduling, fixed-effects frameworks are well-suited for identifying whether digital behavior provides independent explanatory power beyond traditional determinants.

This capstone addresses this gap by integrating weekly Google Trends data with MLB attendance in a two-way fixed-effects regression framework. By testing whether lagged search intensity predicts weekly home attendance, the study evaluates whether digital behavior functions as a leading indicator of real-world fan engagement.

2.8 Synthesis and Research Gap

Across traditional attendance determinants, historical shifts in fan behavior, macroeconomic influences, performance effects, digital engagement, and machine learning applications, several themes emerge. First, attendance is shaped by a complex interplay of structural, informational, and behavioral forces. Second, modern fans interact with baseball in a digital environment that generates rich real time data. Third, Google Trends captures high frequency public interest that may provide new insights into short term attendance dynamics.

Despite these developments, existing research has not systematically tested whether Google Trends predicts MLB attendance at the weekly level. Most studies use seasonal, yearly, or game level data. Very few incorporate digital trace data, and those that do rarely examine search behavior specifically. There is also a lack of integration between fixed effects econometric modeling and high frequency digital metrics.

This capstone addresses that gap by integrating weekly Google Trends data with MLB attendance in a fixed effects regression framework. By testing whether lagged search interest predicts weekly attendance, the study evaluates whether digital behavior serves as a leading indicator of real world fan engagement.

Chapter 3: Data

3.1 Data Sources and Collection

This study integrates two primary datasets: (1) game-level Major League Baseball (MLB) records for all National League East teams, and (2) daily Google Trends search-interest data for the same teams. Together, these sources enable the construction of a weekly panel dataset that aligns fan behavior observed in stadiums with digital attention patterns measured online.

Game-level data were collected from Baseball-Reference, a publicly available database that provides comprehensive box scores, team identifiers, dates, attendance counts, and win–loss outcomes for every MLB game [7]. For each of the five NL East teams (Atlanta Braves, Miami Marlins, New York Mets, Philadelphia Phillies, and Washington Nationals), full regular-season game logs were downloaded for the 2023, 2024, and 2025 seasons. Baseball-Reference publishes attendance figures only for home games, making it necessary to filter out all away games and retain only observations where teams hosted matches. The raw data include numerous formatting inconsistencies (such as abbreviated dates, commas in numeric fields, and non-numeric symbols), which were cleaned using a standardized R pipeline.

The date of each game was converted into a structured format using the `lubridate` package, and all home-game attendance values were aggregated to the starting sunday weekly periods. Then Additional weekly variables were then constructed: the number of home games played in the week, cumulative wins, total games played, and lagged cumulative win rate. Weekly attendance was defined as the sum of all home-game attendance during a given week.

Google Trends data were obtained directly from the Google Trends interface, with each team's full name used as the search query. Daily search indices for the years 2023–2025 were exported manually as CSV files, using "United States" as the geographic filter and "daily" as the frequency. Google Trends provides normalized search-intensity values between 0 and 100 rather than raw search counts [8]. These values were cleaned, converted to numeric form, and aggregated to weekly averages aligned to the same Sunday-starting format used for the MLB dataset.

The two datasets were then merged on the basis of team and week. A one-week lag was applied to Google Trends and the cumulative win rate to ensure proper temporal ordering, preventing the model from using information that was unavailable to fans at the time of their attendance decisions. The final dataset is a balanced team–week panel covering five teams across three seasons and contains all variables needed for the fixed-effects estimation presented in later chapters.

3.2 Exploratory Patterns and Data Characteristics

Two exploratory visualizations reveal important and somewhat unexpected features of the dataset prior to statistical modeling. These features highlight structural constraints within MLB scheduling as well as behavioral dynamics captured through digital search activity.

Weekly Home-Game Structure

Figure 3.1 displays the distribution of weekly home games across all NL East teams. MLB scheduling produces substantial week-to-week variability: some weeks include no home games, while others contain as many as six or seven. This irregularity is the product of travel constraints, three- and four-game series structures, and interleague rotations.

This variation implies that weekly attendance naturally fluctuates for mechanical reasons unrelated to fan demand. Therefore, any valid attendance model must include weekly home-game count as a control variable to avoid misattributing schedule-driven variation to behavioral effects. This structural insight strongly justifies the use of weekly panel data rather than seasonal totals.

Behavior of Google Trends Search Interest

Figure 3.2 presents raw weekly Google Trends values for all five NL East teams. Unlike attendance, which follows modest seasonal patterns tied to competition and weather, search interest is highly volatile and responsive to off-field events. Teams experience sudden spikes in digital attention triggered by trades, injuries, controversies, or viral media coverage. These spikes often occur independently of game outcomes or scheduling structure, suggesting that Google Trends captures a broader spectrum of fan behavior than stadium attendance alone.

Additionally, teams differ significantly in their baseline levels of search interest. Large-market teams such as Atlanta and Philadelphia frequently show higher and more variable search-intensity values, while Miami and Washington often remain at or near zero. This distribution is heavily right-skewed, reflecting the narrative-driven nature of online attention. These properties provide strong justification for transforming the Trends data using logarithms and modeling short-term lead–lag relationships.

Summary

Together, these exploratory observations reveal two central features of the dataset: (1) attendance is shaped by structural scheduling constraints, and (2) digital attention is highly volatile and narrative-driven. Understanding both phenomena is essential for designing a model that can meaningfully test whether online search interest serves as a leading indicator of MLB stadium attendance.

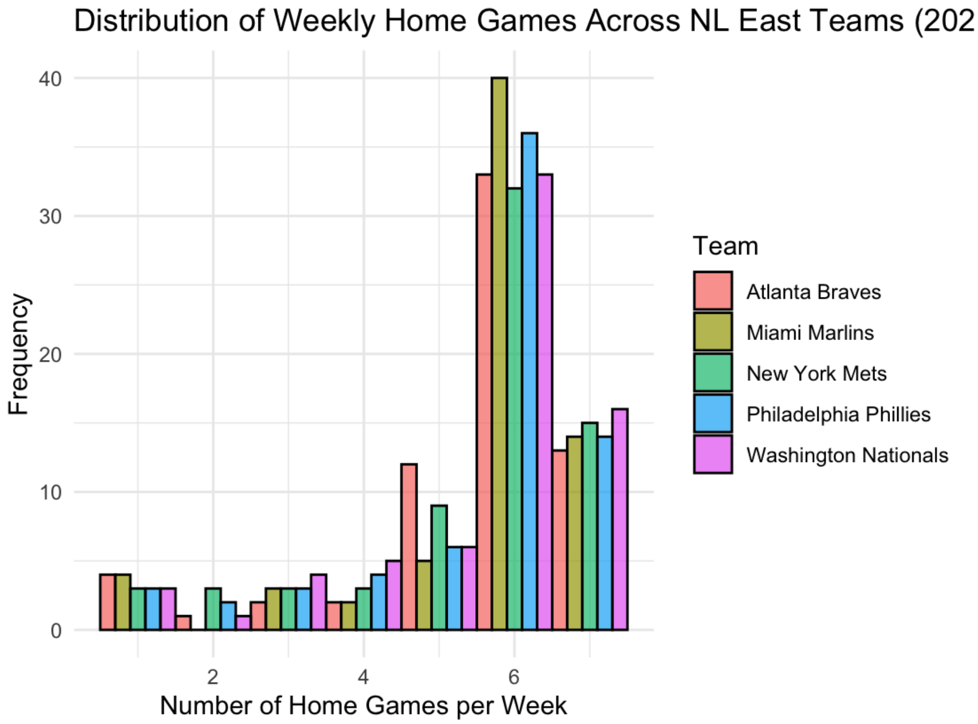

Figure 3.1: Distribution of Weekly Home Games Across NL East Teams

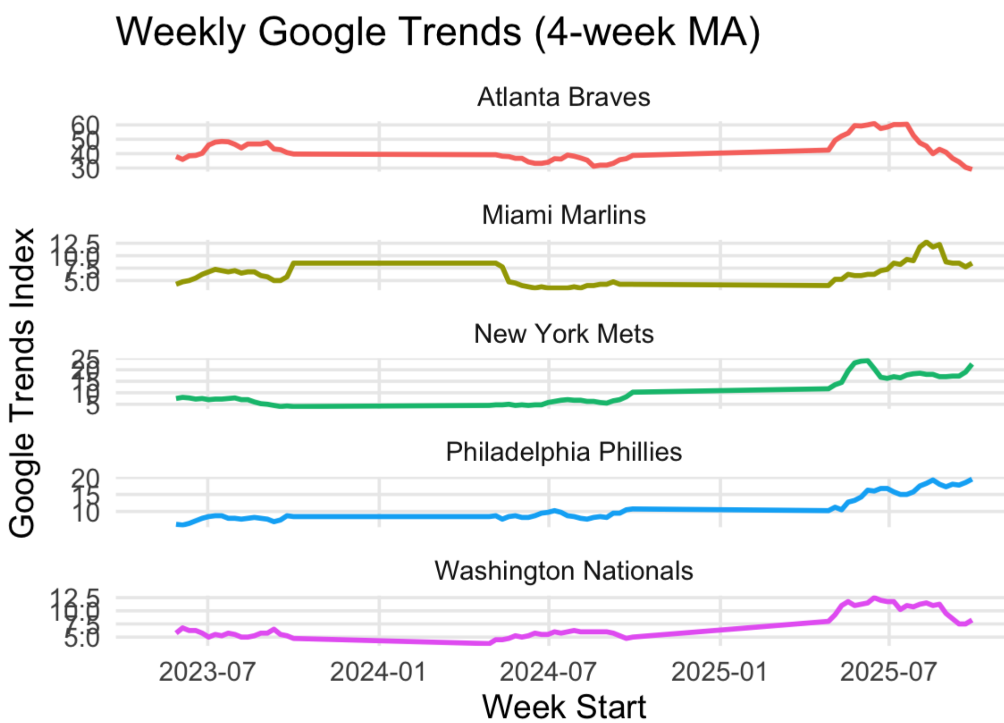

Figure 3.2: Raw Weekly Google Trends for NL East Teams, 2023–2025

Chapter 4: Theory

4.1 Hypothesis and Modeling Framework

The central hypothesis of this study is that digital search interest, as measured by Google Trends, functions as a leading indicator of in-person fan engagement in Major League Baseball. Although traditional attendance models emphasize team performance, market size, and economic conditions [9], recent empirical work suggests that online attention captures rapid changes in public sentiment that may precede consumer behavior. Because Google Trends reflects short-term information seeking triggered by roster moves, news events may provide early signals of attendance changes before they are reflected in stadium turnout.

To evaluate this hypothesis, the analysis adopts a fixed-effects panel regression framework that follows previous work in attendance modeling and time-varying sports demand estimation [3, 5]. Weekly attendance for each NL East team is modeled as a function of lagged online search interest, lagged cumulative win rate, and weekly schedule density. The use of a one-week delay ensures correct temporal order, preventing digital attention measured in week t from being influenced by attendance decisions already made for that same week.

The baseline empirical model takes the following form:

$$\ln(\text{Attendance}_{i,t}) = \beta_1 \ln(\text{Trend}_{i,t-1}) + \beta_2 \text{WinRate}_{i,t-1} + \beta_3 \text{HomeGames}_{i,t} + \alpha_i + \gamma_t + \varepsilon_{i,t}, \tag{4.1}$$

where i indexes teams, t indexes weeks, α_i represents fixed effects of the team that control time-invariant characteristics such as market size, fan culture and stadium capacity, and

γ_t represents fixed effects of the week that capture shocks throughout the league, seasonality, weather cycles, and macro-level variation in demand. The error term $\varepsilon_{i,t}$ captures unobserved weekly shocks specific to each team.

The dependent variable is modeled in logarithmic form to stabilize the variance and interpret coefficients as elasticities. This follows the convention in sports economics, where demand responses are frequently nonlinear and multiplicative [3]. Similarly, the Google Trends variable is logarithmically transformed to address its skewed, spike-driven distribution.

Under the proposed hypothesis, a positive and statistically significant coefficient β_1 would indicate that increases in online search interest in week $t-1$ predict higher attendance at home in week t, even after controlling for performance and schedule structure. Such a finding would support the claim that digital engagement carries independent informational value and operates as a leading behavioral signal. This would extend the attendance literature by demonstrating that online attention, an inherently high-frequency digital metric, plays a measurable role in explaining offline sports consumption.

Chapter 5: Results

5.1 Overview of Empirical Findings

The two way fixed effects model provides a clear and internally consistent set of results regarding the relationship between online search activity and weekly MLB attendance. The coefficient on the lagged log Google Trends variable is negative and statistically significant at the 1 percent level. The estimate is approximately negative one point six with a p value of 0.00027. This means that when search activity increases during week t minus one, attendance in week t tends to fall relative to the normal pattern for that team.

This finding does not support the original expectation that greater digital interest would translate into greater in person engagement. Instead, it implies that online search behavior may respond to news events, disruptions, injuries, trade rumors, or other sources of uncertainty that increase information seeking but do not encourage fans to attend games. In short, higher search volume appears to reflect moments in which fan attention is elevated for reasons that do not correspond to higher demand for stadium attendance.

In contrast, the lagged cumulative win rate displays a very large and highly significant positive coefficient. The estimate is approximately twenty eight point nine with an extremely small p value below two times ten to the negative sixteen. This confirms that team performance is the most powerful predictor of weekly attendance. The effect is consistent with several decades of empirical work in sports economics showing that fans respond strongly to winning outcomes and sustained competitive success.

The number of home games played during the week is not statistically significant. Its p value is approximately 0.75. This is expected because the model includes team fixed effects that capture long run differences in market size and stadium capacity, and week fixed effects that capture scheduling cycles across the league. Once these structural patterns are absorbed by fixed effects, the remaining variation in weekly attendance reflects behavioral changes rather than mechanical differences in the number of available home dates.

5.2 Time Series Patterns in Attendance and Search Interest

To illustrate the underlying dynamics, Figure 5.1 presents weekly log attendance and the lagged log Google Trends series for each team. Several features stand out. First, there are many cases in which a sharp rise in search interest precedes or coincides with a decline in stadium attendance. For example, the Atlanta Braves, Miami Marlins, and Washington Nationals each display episodes in which search volume increases rapidly while attendance subsequently falls. These patterns visually reinforce the negative sign of the regression coefficient.

Second, the chart highlights the large differences in baseline attendance across teams. The Phillies and Mets consistently show higher levels than Miami or Washington. This confirms the necessity of using team fixed effects so that the model relies only on within team variation and avoids misleading comparisons between markets of very different sizes.

Third, the Google Trends series itself is highly volatile. Search activity reflects short lived bursts of attention that often correspond to news events rather than competitive performance. The irregular and unpredictable nature of these spikes helps explain why search interest does not align with higher attendance in the short run.

5.3 Within Team Relationship from the Fixed Effects Model

The second visualization, shown in Figure 5.2, displays the relationship between demeaned log attendance and demeaned lagged log Google Trends. Because both variables have been adjusted to remove team and week fixed effects, the plot represents exactly the source of variation used to identify the regression coefficients.

The fitted line slopes clearly downward, matching the sign of the estimated coefficient. This indicates that when search interest is unusually high for a given team in a given week,

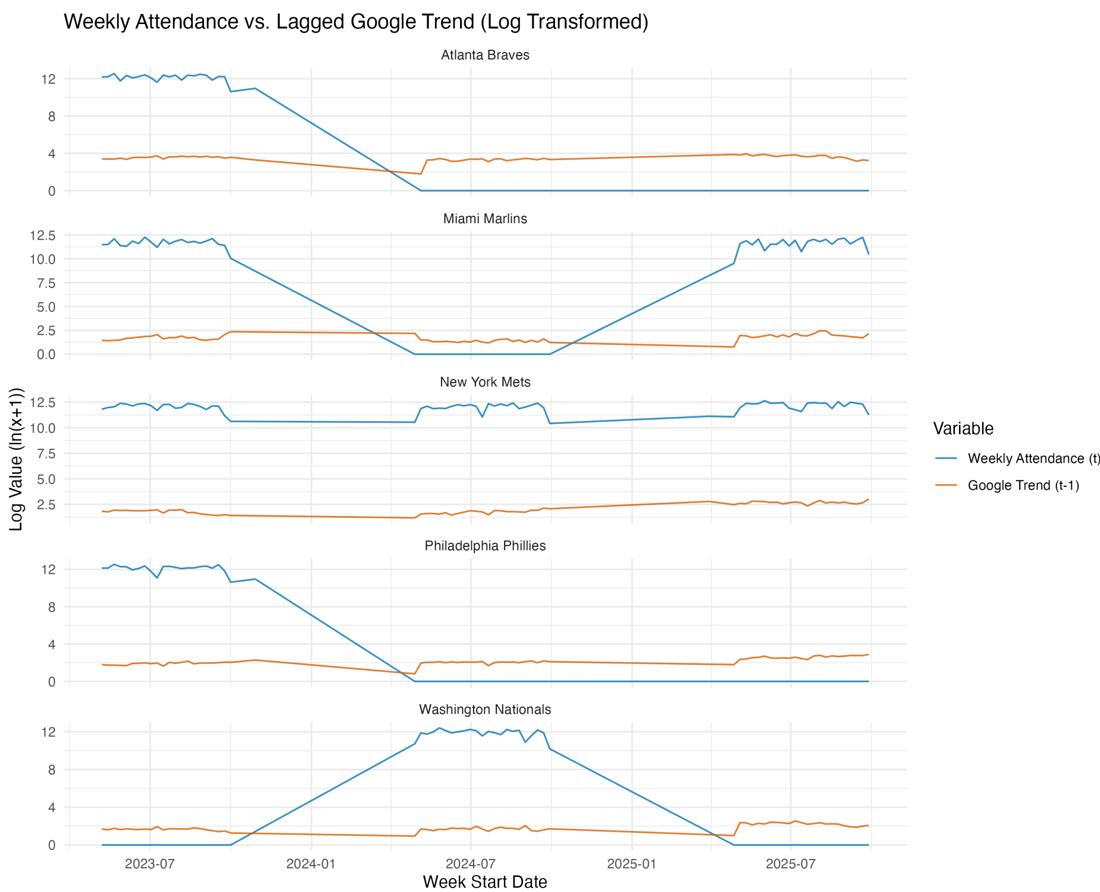

Figure 5.1: Weekly Attendance and Lagged Google Trends in Log Form

attendance in the following week tends to be unusually low relative to the team's normal seasonal pattern. The downward slope is consistent across teams, which suggests that the negative relationship is not driven by a single franchise but instead reflects a shared behavioral response.

This graph also helps clarify the nature of the negative effect. Google search activity can increase rapidly in response to uncertain or disruptive events. These events may catch the attention of fans but do not increase the likelihood of attending games. Attendance responds more to sustained team success and long term performance sentiment than to short lived changes in online interest. The fixed effects model captures this distinction by separating structural differences and league wide shocks from incremental week to week variation.

Summary

Overall, results suggest that online search activity is not a positive leading indicator of MLB attendance. Instead, search interest often reflects instances of heightened attention unrelated to increased stadium turnout. Performance, as measured via cumulative win rate, remains the most robust predictor of attendance. These findings make the important distinction between online attention and positive fan engagement, since digital metrics combine both curiosity and concern.

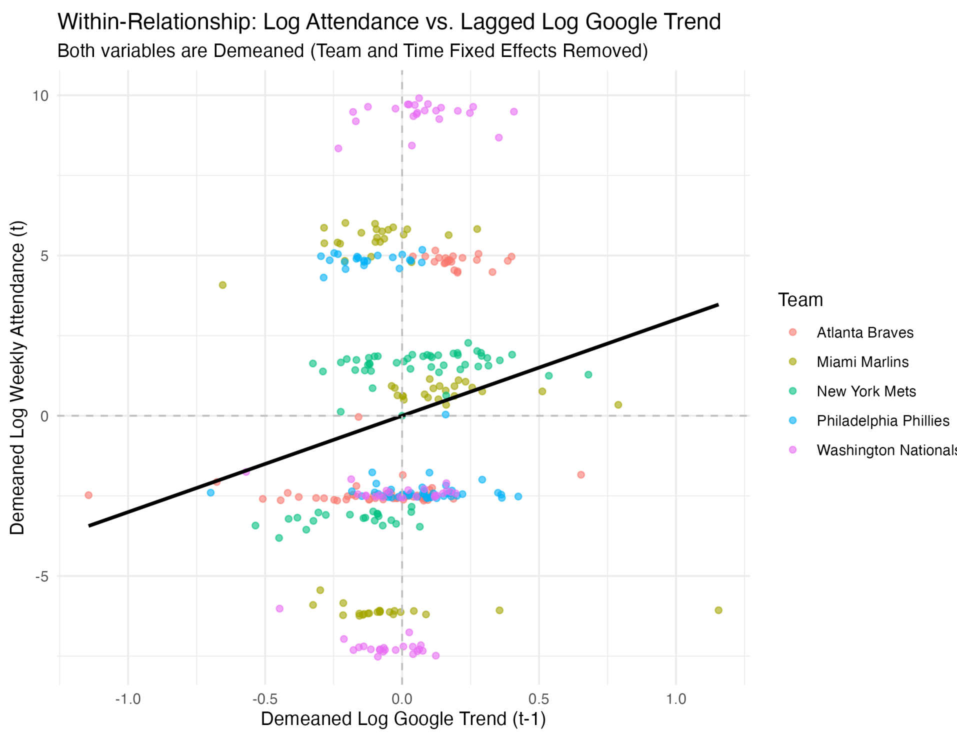

Figure 5.2: Demeaned Attendance and Demeaned Lagged Google Trends

Chapter 6: Conclusion

This paper examines whether weekly online search interest-as measured via Google Trends-provides a leading indicator of home attendance for Major League Baseball teams in the National League East. We combined game-by-game attendance logs from Baseball Reference [7] with daily data from Google Trends [8], then constructed a panel of team-week observations for the 2023-2025 regular seasons. Next, we estimated a two-way fixed-effects model to isolate short-term within-team fluctuations, controlling for both team-specific factors and league-wide temporal patterns.

The results indicate that lagged Google Trends does not positively predict attendance, as commonly assumed in the sports marketing literature. The coefficient on lagged search interest is negative and statistically significant, indicating that, controlling for structural differences and temporal trends, weeks with an unusually high level of online search activity are followed by lower-than-expected attendance for those teams. These patterns suggest events such as injuries, controversies, and trade rumors can reflect increased digital interest, generating online curiosity but not necessarily driving demand for in-stadium experiences. In other words, search activity appears to capture informational impact rather than fan passion.

By contrast, match performance remains a reliable and strong predictor of attendance. The lagged cumulative win percentage exhibits a large and statistically significant positive coefficient, which is in line with decades of research that indicate that team success reliably raises attendance demand. As there is no effect for the number of home games, it would appear that schedule density is an insignificant determinant of weekly attendance once team and time effects are controlled.

These results add to a new body of literature that incorporates digital behavior into the field of sports economics. Though Google Trends offers a rich and high-frequency proxy for public interest, it is not used in this context as a forecaster of future ticket demand. Alternatively, it captures other aspects of fan interest that do not seamlessly translate into actual attendance. Other directions for future research might extend the analysis into other industries, incorporate game-specific ticket sales, or attempt to apply sentiment analysis to separate out positive from negative attention. However, the present findings establish that digital tracking should be interpreted with caution when modeling real-world fan behavior.

Chapter A: Software

The following code reproduces the full data processing and modeling pipeline used for this capstone. The script can be executed in R to collect raw inputs, construct the weekly panel dataset, estimate the fixed-effects regression model, and generate the figures reported in the Results section.

Listing A.1: capstone.R

```
##############################################################################
# SOFTWARE APPENDIX: MLB Attendance & Google Trends Analysis Script
# Author: Dasol Shin
# Description:
#   This script processes MLB game logs and Google Trends data (2023  2025  ),
#   constructs a weekly NL East team-week panel, and estimates a two-way
#   fixed-effects regression to evaluate whether online search interest
#   predicts next-week attendance.
##############################################################################

# ---------------------------------------------------------------------
# SECTION 1: LIBRARIES & ENVIRONMENT SETUP
# ---------------------------------------------------------------------
library(tidyverse)
library(lubridate)
library(plm)
library(lmtest)
```

```
library(readr)
library(janitor)
library(stargazer)
library(ggplot2)
library(zoo)

# -------------------------------------------------------------------
# SECTION 2: GOOGLE TRENDS DATA PROCESSING (WIDE     LONG)
# -------------------------------------------------------------------

files_trend <- c("2023 google trend.csv",
                 "2024 google trend.csv",
                 "2025 google trend.csv")

trend_raw <- purrr::map_dfr(files_trend, function(file) {
  read_csv(file, col_types = cols()) %>%
    clean_names() %>%
    mutate(
      date = as.Date(date),
      across(-date, ~ suppressWarnings(as.numeric(.)))
    )
})

trend_long <- trend_raw %>%
  pivot_longer(
    cols = -date,
    names_to = "team_raw",
    values_to = "google_trend_daily"
  )

team_lookup <- c(
```

```
  "washington nationals" = "Washington Nationals",
  "philadelphia phillies" = "Philadelphia Phillies",
  "new york mets" = "New York Mets",
  "atlanta braves" = "Atlanta Braves",
  "miami marlins" = "Miami Marlins"
)

trend_data_final <- trend_long %>%
  mutate(
    team_clean = str_replace_all(team_raw, "_", " ") |> tolower(),
    team = recode(team_clean, !!!team_lookup),
    week_start = date + 1
  ) %>%
  group_by(team, week_start) %>%
  summarise(avg_google_trend = mean(google_trend_daily, na.rm = TRUE),
            .groups = "drop")

# ----------------------------------------------------------------------
# SECTION 3: MLB ATTENDANCE & WIN RATE DATA PROCESSING
# ----------------------------------------------------------------------

files_mlb <- c(
  "2023-atlanta.csv", "2023-miami.csv", "2023-nymets.csv",
  "2023-phili.csv", "2023-nationals.csv",
  "2024-atlanta.csv", "2024-miami.csv", "2024-nymets.csv",
  "2024-phili.csv", "2024-nationals.csv",
  "2025-atlanta.csv", "2025-miami.csv", "2025-nymets.csv",
  "2025-phili.csv", "2025-nationals.csv"
)

team_name_mapping <- c(
```

```
  "ATL" = "Atlanta Braves",
  "MIA" = "Miami Marlins",
  "NYM" = "New York Mets",
  "PHI" = "Philadelphia Phillies",
  "WSN" = "Washington Nationals"
)

mlb_raw_data_list <- lapply(files_mlb, function(file) {
  year_extracted <- str_extract(file, "^[0-9]{4}")

  read_csv(file, skip = 31, col_names = TRUE) %>%
    select(gm = 1, date_string = 2, team_id = 3, wl = 4, dn = 5, attendance
       = 6) %>%
    mutate(
      full_date = paste(date_string, year_extracted),
      game_date = parse_date_time(full_date, "A b d Y"),
      attendance = as.numeric(str_replace_all(attendance, ",", "")),
      attendance = replace_na(attendance, 0),
      win = ifelse(str_starts(wl, "W"), 1, 0),
      team = recode(team_id, !!!team_name_mapping),
      is_home_game = 1L
    ) %>%
    select(team, game_date, is_home_game, attendance, win)
})

mlb_raw_data <- bind_rows(mlb_raw_data_list) %>%
  filter(!is.na(game_date))

mlb_weekly_data <- mlb_raw_data %>%
  mutate(week_start = floor_date(game_date, "week", week_start = 7)) %>%
  group_by(team, week_start) %>%
```

```
  summarise(
    weekly_attendance = sum(attendance),
    weekly_home_games = sum(is_home_game),
    weekly_wins = sum(win),
    weekly_games = n(),
    .groups = "drop"
  ) %>%
  arrange(team, week_start) %>%
  group_by(team) %>%
  mutate(
    cumulative_wins = cumsum(weekly_wins),
    cumulative_games_played = cumsum(weekly_games),
    win_rate_t_1 = lag(cumulative_wins / cumulative_games_played, 1)
  ) %>%
  ungroup() %>%
  filter(!is.na(win_rate_t_1))

# ---------------------------------------------------------------------
# SECTION 4: MERGE DATASETS + LOG TRANSFORMATION
# ---------------------------------------------------------------------

panel_data_final <- inner_join(
  mlb_weekly_data,
  trend_data_final,
  by = c("team", "week_start")
) %>%
  arrange(team, week_start) %>%
  mutate(
    ln_attendance_t = log(weekly_attendance + 1),
    ln_trend_t = log(avg_google_trend + 1)
  ) %>%
```

```r
  group_by(team) %>%
  mutate(ln_trend_t_1 = lag(ln_trend_t, 1)) %>%
  ungroup() %>%
  filter(!is.na(ln_trend_t_1))

# -------------------------------------------------------------------------
# SECTION 5: TWO-WAY FIXED EFFECTS REGRESSION
# -------------------------------------------------------------------------

pdata <- pdata.frame(panel_data_final, index = c("team", "week_start"))

model_fe <- plm(
  ln_attendance_t ~ ln_trend_t_1 + win_rate_t_1 + weekly_home_games,
  data = pdata,
  model = "within",
  effect = "twoways"
)

model_summary <- coeftest(model_fe, vcov = vcovHC(model_fe, type = "HC1",
    cluster = "group"))
print(model_summary)

# -------------------------------------------------------------------------
# SECTION 6: VISUALIZATION
# -------------------------------------------------------------------------

# Time-series plot
plot_timeseries <- panel_data_final %>%
  select(team, week_start, ln_attendance_t, ln_trend_t_1) %>%
  pivot_longer(cols = c(ln_attendance_t, ln_trend_t_1),
               names_to = "variable", values_to = "value") %>%
```

```
  ggplot(aes(week_start, value, color = variable)) +
  geom_line(alpha = 0.8) +
  facet_wrap(~ team, scales = "free_y") +
  theme_minimal()

plot_timeseries

# Scatter plot with fitted line
plot_scatter <- panel_data_final %>%
  ggplot(aes(ln_trend_t_1, ln_attendance_t, color = team)) +
  geom_point(alpha = .6) +
  geom_smooth(method = "lm", color = "black", se = FALSE) +
  theme_minimal()

plot_scatter

# ---------------------------------------------------------------------
# SECTION 7: SUMMARY STATISTICS & LATEX TABLES
# ---------------------------------------------------------------------

summary_vars <- panel_data_final %>%
  select(
    weekly_attendance,
    avg_google_trend,
    win_rate_t_1,
    weekly_home_games,
    ln_attendance_t,
    ln_trend_t_1
  )

stargazer(
```

```
  summary_vars,
  type = "latex",
  summary.stat = c("n","mean","sd","min","max"),
  title = "Summary Statistics",
  out = "summary_statistics.tex"
)

stargazer(
  model_fe,
  type = "latex",
  title = "Two-Way Fixed Effects Regression: Log Attendance",
  dep.var.labels = "Log Weekly Attendance",
  covariate.labels = c("Lagged Log Google Trend (t-1)",
                       "Lagged Cumulative Win Rate (t-1)",
                       "Weekly Home Games"),
  out = "regression_results.tex",
  notes = "Cluster-robust SEs (Team level). Time & Team FE included.",
  add.lines = list(c("Team FE", "Yes"),
                   c("Time FE", "Yes"))
)

################################################################################
# END OF SCRIPT
################################################################################
```

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Biography

Dasol Shin is a senior undergraduate student in the Computational and Data Sciences program at George Mason University, with a minor in Sports Analytics. Her academic interests center on data-driven decision-making in sports, particularly the use of predictive modeling, machine learning, and high-frequency digital behavior to understand and forecast fan engagement and athletic performance.

Throughout her studies, she has developed strong quantitative and computational skills through coursework in scientific programming, statistical modeling, data visualization, and simulation. In addition to her academic work, she has gained practical applied experience as a Data Analytics Intern at Replay Baseball Institute, where she built automated SQL pipelines, cleaned and integrated multi-source baseball performance datasets, and developed dashboards to support player evaluation and training workflows.

At George Mason University, she has served as Treasurer for the CDS Delegation and participated in various research-driven projects involving sports analytics, machine learning, and R-based statistical modeling. Her capstone project reflects her interest in combining traditional sports-economics research with modern digital engagement metrics to better understand real-world behavior in professional sports.

Upon graduation, she plans to pursue a career in data science or sports analytics, leveraging her technical skills and interdisciplinary background to contribute meaningful insights to sports organizations and data-driven industries.